

“Amazing Sketchbook the Ride”: Driving a Cart in a 3DCG Scene Created from a Hand-Drawn Sketch

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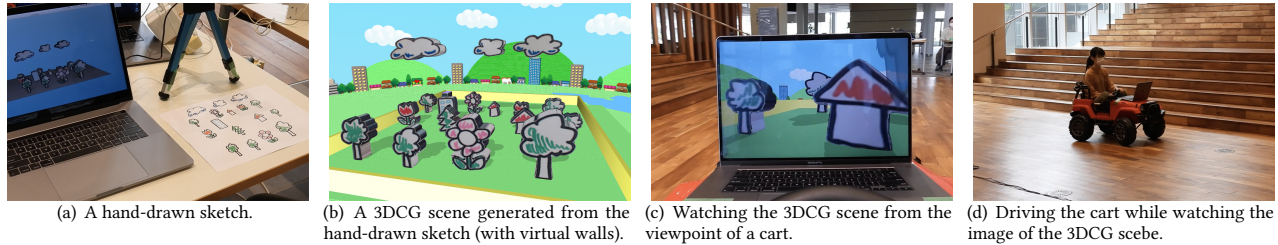


Figure 1: Overview of “Amazing Sketchbook the Ride”.

CCS CONCEPTS

• Computing methodologies → Computer graphics; • Human-centered computing → Human computer interaction (HCI).

KEYWORDS

Sketch, 3DCG, Driving, Cart, Immersion, Interaction

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1 INTRODUCTION

In this research, we introduce an interactive content “Amazing Sketchbook the Ride”, which allows users to drive an actual cart and move freely in a 3DCG scene as if they were immersed in it. Although some similar contents have already been developed [MasterofShapes 2018][Nintendo 2020][Motoyama 2018], the 3DCG scene that can be moved was generated in advance, or CG was synthesized into the real space by AR technology. On the other hand, the users can create the 3DCG scene for driving by themselves by drawing a sketch in our content.

First, the user draws a sketch of a town on a paper with a pen (Figure 1(a)). Next, a 3DCG scene of the town is automatically generated from the sketch by taking it with a camera (Figure 1(b)). Then, the user rides on a cart equipped with a display in which the 3DCG scene is displayed (Figure 1(c)). While watching the

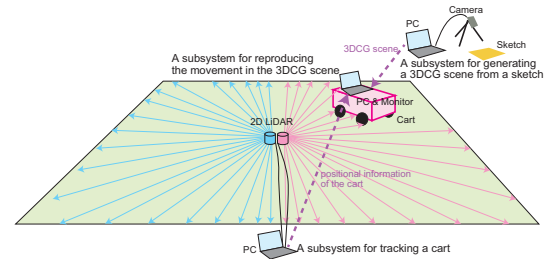


Figure 2: The system for “Amazing Sketchbook the Ride”.

3DCG scene in the display, the user can actually drive the cart and move in the 3DCG scene of the town created by him/herself (Figure 1(d)). Interactions such as destroying CG objects in the 3DCG scene according to the movement of the cart are also possible.

2 METHODS

In this content, the user first draws a sketch to generate a 3DCG scene, and then drives a cart to move through the generated 3DCG scene. The system consists of three subsystems: for generating a 3DCG scene from a sketch, for tracking a cart, and for reproducing the movement in the 3DCG scene (Figure 2). The subsystems are connected by a network each other.

2.1 Generating a 3DCG scene from a sketch

The subsystem for generating a 3DCG scene from a sketch consists of a PC and a camera. The method of generating a 3DCG scene from a sketch is based on our previous study, “Amazing Sketchbook”, which can automatically generate a 3DCG scene by image analysis of a sketch taken with a camera and generating CG objects from hand-drawn objects in the sketch [Mizuno and Funahashi 2017].

In the former method, the CG objects in the 3DCG scene were flat. In the content of this research, the user can move freely in the 3DCG scene, and the CG objects can be observed from various directions. Therefore, we improved our method so that the CG objects become three-dimensional by making them thicker in this research (Figure

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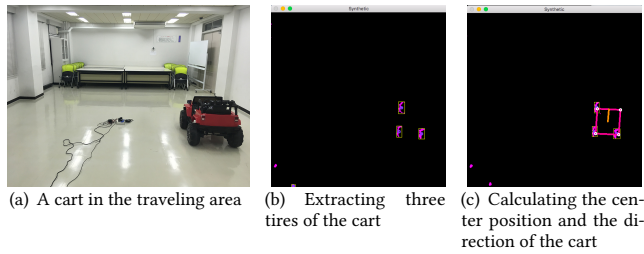


Figure 3: Tracking a cart.

1(a)(b)). The thickness of each object is automatically determined based on its width.

After the 3DCG scene is generated, the data is transferred to the subsystem for reproducing the movement in the 3DCG scene using a file sharing function.

2.2 Tracking a cart

The subsystem for tracking a cart consists of a PC and two 2D LiDARs. The LiDARs are installed in the center of the area where the cart travels, and the floor of the whole area is scanned two-dimensionally at 30 (fps) with an angular resolution of 0.125 (deg).

The target of scanning is four tires of the cart (Figure 3(a)). The scan results from the two 2D LiDARs are integrated and obtained as a single binary image with 1 (pixel) = 1 (cm). When the cart is in the area, the tires of the cart are extracted as two to four small areas. Then, the center of gravity of each area is set as the position of each tire (Figure 3(b)).

When the positions of four tires are obtained, each tire is first manually labeled in front left, front right, back left, and back right. After manually labeling each tire, then each tire is automatically labeled in consideration of the amount of movement of each tire between frames, and the position is tracked. The size of the cart and the positional relationship of the four tires are known. Therefore, if at least two or more tire positions can be obtained and labeled, the positions of other tires can be calculated. By acquiring the positions of the four tires of the cart, the center position and the direction of the cart can be calculated (Figure 3(c)).

The positional information of the cart is sent to the subsystem for reproducing the movement in the 3DCG scene via the network using the UDP protocol at 30 (fps).

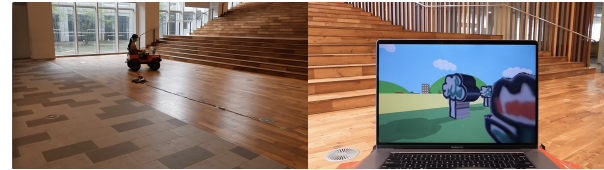
2.3 Reproducing the movement of the cart in the 3DCG scene

The subsystem for reproducing the movement of the cart in the 3DCG scene consists of a PC and a monitor mounted on the cart. As mentioned in Section 2.1, the subsystem has received the data and constructed the 3DCG scene. Then the subsystem receives the positional information of the cart as mentioned in Section 2.2, applies it to the viewpoint in the 3DCG scene, and renders an image of the 3DCG scene. Virtual walls are automatically generated in the 3DCG scene corresponding to the area where the cart can move.

By receiving the position information of the cart and rendering the 3DCG scene sequentially, the image on the monitor changes in real time as the cart moves. When driving the cart while watching the image on the monitor, the user can feel as if moving in the 3DCG scene created from the hand-drawn sketch with the cart.



(a) The cart in the traveling area and the 3DCG scene image on the monitor



(b) The cart moving in the area and the 3DCG scene image on the monitor



(c) The cart hitting a 3DCG object in the 3DCG scene

Figure 4: Experiment with a cart for children.

The system measures the distance between each CG object in the 3DCG scene and the position information of the cart, and it is possible to determine if the cart collides with them. This information is used to interact with the 3DCG scene via the cart, such as destroying 3DCG objects by hitting them with the cart.

3 EXPERIMENTAL RESULTS

We implemented the system and conducted experiments. The cart used in the experiment is a toy for children (max speed: 5 (km/h)) and the area for driving is a rectangular area of 9×7 (m).

The position and the direction of the cart could be obtained correctly by the proposed method. The viewpoint was set correctly in the 3DCG scene generated from the sketch, and the 3DCG scene seen from the cart was properly displayed on the monitor attached to the cart (Figure 4(a)). When the cart started to move, the 3DCG scene displayed on the monitor attached to the cart changed smoothly according to the movement of the cart (Figure 4(b)). When the cart hits an object in the 3DCG scene while driving, the object was destroyed (Figure 4(c)).

It was possible to drive the cart while watching only the image of the 3DCG scene on the monitor. The video delay with respect to the movement of the cart was about 0.1 (sec). The test user did not feel discomfort when driving the cart, and felt immersed in the 3DCG scene generated from the hand-drawn sketch while driving.

We realized a new experience of freely driving on a cart in the 3DCG scene generated by our hand-drawn sketch. Realizing more diverse interactions during driving is one of future works.

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